

IG AGENT

v25 — Complete Final Specification

The definitive north star for the entire v25 build
Points Reinforcement | Three-Speed Learning | Web Dashboard | ATR Exits | 8 Weeks to Live

Version	v25-dev
Foundation	v24-proven
Date	May 2026
Target	8 Weeks
Status	FINAL v7

Purpose — Read This First

This is the complete, final specification for IG Agent v25. It incorporates every design decision, every lesson learned from v24's proven trading sessions on 25 May 2026, and every enhancement identified through continued Cursor development. v25 is a step change across three dimensions: (1) a complete GUI rebuild from Tkinter to a web-based dashboard, (2) a full AI/ML learning layer with points-based reinforcement, and (3) a production-grade trading engine with all operational fixes proven in v24.

This document is Cursor's north star. It supersedes ALL previous versions (v1–v6).

v7 update: Incorporates all additions and fixes from Cursor development since v6, including Blocks A–F reliability controls, historical data ingestion pipeline, ML training infrastructure, Dashboard Intelligence tab, client sentiment, and all operational constraints. Preserve existing gate numbering (7 gates in risk validation, 13 gates total), Japan 225 epic, and DEMO/LIVE modes. ML and replay scheduler are operational subsystems — not required for live trading.

1. What v25 Is — The Step Change

v25 is Three Step Changes in One

STEP 1: GUI — Replace Tkinter with a web-based dashboard. Professional real-time trading interface accessible from any device. STEP 2: AI/ML — Points-based reinforcement learning that scores every trade, learns from wins and losses, and continuously improves signal quality. STEP 3: Production Trading Engine — All operational fixes proven on 25 May 2026 built in from day one.

Dimension	v24 Proven	v25 Target
GUI Framework	Tkinter — beach ball on every tab	Web dashboard — React + FastAPI, no freeze ever
Stream	Lightstreamer (fixed 25/05)	Lightstreamer + WebSocket to dashboard
Signal gate	Confidence ≥ 92 only	Conf + ML + fitness + points — all agree
Exit management	Fixed stop + fixed target	ATR trail + break-even + partial close
Learning	None	Points reinforcement + nightly 100x replay
Trade data integrity	Phantom trades, wrong currency (fixed 25/05)	IG-confirmed only, GBP always, pending state
Session management	Manual restart at market open (fixed 25/05)	Automatic session refresh, gap protection
Order execution	Synchronous — blocked loop (fixed 25/05)	Async — loop never blocked
Position sync	Stopped on watchdog kill (fixed 25/05)	Independent of loop lifecycle
Watchdog	Killed at 60s during order confirm (fixed 25/05)	300s + 3 auto-restarts
REST budget	Saturated at open (fixed 25/05)	Protected at open, reserved during orders
Deal reconciliation	Open/close ID mismatch (fixed 25/05)	Dual ID matching, auto-reconciliation script
Win rate target	55% all-time (proven)	60%+ consistently
Monitoring	Tab-based, beach ball, manual	Real-time web dashboard, mobile accessible

2. Lessons Learned from v24 — Built Into v25 From Day One

What 25 May 2026 Taught Us

Every fix made on 25 May exposed a design decision that v25 must get right from the start.

Lesson	v24 Problem	v25 Solution
Lightstreamer auth	PROXY_SERVER adapter rejected; CST as username instead of account_id	adapter=None, setUser(account_id), CST- XST- password — correct from day one
Market open session	Afternoon session carried to evening; REST saturated; stream got zero ticks	Auto session refresh on market open, 120s REST pause, stream gets priority
Async orders	Synchronous REST order blocked the trading loop for 20-40s	All orders async via background worker thread — loop never blocked
Watchdog false kills	65s threshold killed loop during order confirm	300s tick-count watchdog + 3 auto-restarts
Position sync lifecycle	Stopped when loop died; ghost positions overnight	Position sync independent of loop — always runs while DEMO session active
Quote freshness	LS CONNECTED but hub stale; _next_quote skipped REST fallback	Direct hub.publish() in onItemUpdate; REST fallback when age > 35s regardless
Deal ID mismatch	Open deal ID (DIAAA) never matched close ref (LBH6) in txn sync	Store both IDs; reconcile via dual lookup; auto-reconciliation script
Currency display	pnl_points shown as \$USD instead of GBP	Account currency from IG session always; pts shown as pts when unconfirmed
Phantom trades	Local position-sync closes shown as confirmed before IG confirms	Pending state until IG transaction confirms; never show WIN/LOSS until confirmed
SIM/soak pollution	179 SIM trades could leak into main display	EXCLUDED_SOURCES permanent filter; SIM never shown in main GUI
LiveTradeGate	5 arming ticks blocked first signal; WAIT->BUY edge requirement	2 arming ticks; any validated signal after arming passes
GUI beach ball	Tkinter renders lifecycle diagram on main thread; 3-4s freeze	Web dashboard — React renders in browser; zero main thread blocking ever
Transaction sync at open	force=True startup sync saturated REST budget	120s defer when market open at launch; paused during order in flight

Lesson	v24 Problem	v25 Solution
Signal threshold	92 too tight for proof; 90 used in practice	Review threshold after 20 sessions; document in config clearly

3. Three-Speed Learning System

3.1 Nightly Replay Cycle

Without replay: Demo accumulates 200-300 labels over 8 weeks. With replay: first overnight run generates 2,000+ labels from 2 years of real Nikkei history. By Week 4 the model has 8,000+ labels. The agent arrives at Demo already trained.

Speed / Mode	Purpose
100x — Nightly replay (TEST)	Bulk training from 2 years real Nikkei 5m/15m history. Runs nightly 06:15–22:30 BST. Generates thousands of labelled decisions overnight. Model improves every night. Operational subsystem — not required for live trading.
1x — Shadow (DEMO)	Validates replay model on real live ticks. Logs what ML would decide alongside rules-only trading. Confirms model generalises beyond history.
1x — Live trading (DEMO/LIVE)	Ground-truth labels — 3x weight. Every real trade outcome feeds back. Continuously improves model quality.

Nightly Replay Schedule

Time	Action
06:00 BST	Japan 225 session closes — session end flatten runs
06:15 BST	Replay scheduler starts automatically (scripts/replay_scheduler.py)
06:15–08:00	Pull new IG bars, run 100x replay, extract labels, retrain model
08:00–09:00	Evaluate vs current model; stage for approval if better
22:30 BST HARD CUTOFF	Replay stops — live trading gets full REST budget. No run 22:30–07:00.
23:11 BST	Japan 225 opens — agent trades with latest approved model

3.2 Label Weighting

Source	Weight & Rationale
Real DEMO trade WIN/LOSS	3x — Ground truth: agent actually executed
Replay label WIN/LOSS	1x — Simulated on real history: good signal, lower trust
Shadow label (would-have)	0.5x — Indirect inference: ML predicted, not confirmed

4. Points-Based Reinforcement System

4.1 Scoring Formula

Points are confidence-weighted — a high-confidence loss is the most damaging event. Points drive position sizing, trade frequency, and automatic backing-off.

Event	Points
WIN — high conviction (ML ≥ 0.85)	+3 x (actual_pnl / avg_win_pnl)
WIN — standard (ML 0.65–0.85)	+2 x (actual_pnl / avg_win_pnl)
WIN — marginal (ML at threshold)	+1 flat
LOSS — high conviction (ML ≥ 0.85)	-4 x (actual_loss / avg_loss) — LARGEST PENALTY
LOSS — standard (ML 0.65–0.85)	-2 x (actual_loss / avg_loss)
LOSS — marginal	-1 flat
BREAKEVEN	0
Partial close — banked portion	Scored immediately on banked P&L
Trail exit better than fixed would have been	+0.5 bonus
ML suppressed — would have lost	+0.5 (capital protected)
ML suppressed — would have won	-0.5 (missed opportunity)

4.2 Staged Backing-Off and Recovery

State	Cumulative Points	Behaviour	Recovery
HEALTHY	> +10	Full size, standard threshold, max 6 positions	N/A
CAUTION	-5 to +10	0.75x size, threshold +2, max 3 positions	Return to +10 OR 5 wins
WARNING	-15 to -5	0.5x size, threshold +5, max 2 positions	Return to -5 OR 3 wins
DANGER	-30 to -15	0.25x size, threshold +10, max 1 position	Return to -15 OR 3 wins
STOP	< -30	No new trades — manual approval required	Manual only
SESSION PAUSE	3 consec losses	Skip next 3 signals	After 3 skips

State	Cumulative Points	Behaviour	Recovery
DAY STOP	Session score < -5	No more trades today	Next session open

5. Exit Management — ATR Trailing Stops

Fixed stops cap upside and ignore momentum. ATR trailing stops let winners run while protecting capital. Partial closes bank profit while holding a runner. This directly grows the points score by maximising winning trade value.

Exit Mechanism	Logic
Break-even trigger	Trade reaches +1.0x ATR profit → stop moves to entry price
Trail activation	Immediately after break-even — follows price in profit direction only
Trail distance — high ML (>= 0.85)	1.75x ATR — high conviction gets more room to run
Trail distance — standard	1.5x ATR
Trail distance — low ML (< 0.70)	1.0x ATR — tighter on weaker signals
Partial close trigger	Trade reaches +1.5x ATR profit → close 50%, score points immediately
Momentum tighten	RSI diverges against trade → trail tightens to 0.75x ATR
Time-based exit	Open > 6 bars AND price moved < 0.3x ATR → exit at market
Maximum cap	3.0x ATR profit — hard exit regardless of trail
Re-entry after trail	15m trend still strong → re-entry eligible at threshold +3, 0.75x size
Short trades	Identical logic — mirrored direction
Iron rule	Trail ONLY moves in profit direction — NEVER backwards

6. Session Management — Proven Fixes Built In

Requirement	Implementation
Auto session refresh on market open	Detect closed->open transition; stop stale session; start fresh DEMO automatically
REST budget protected at open	120s pause on background REST; stream and stream probe get priority
Transaction sync deferred at open	force=True startup sync waits 120s when market already open at launch
REST budget reserved during orders	begin_order_in_flight() pauses non-essential REST until confirm returns
Session end exit (mandatory)	All positions closed T-5min before session end — automatic, no exceptions
Gap open protection	Fitness capped at 40 for first 15min after gap > 1.0x ATR
Cold start protection	Fitness capped for first 6 bars — agent needs context before signals
DST-aware session times	Session times from IG market details API — never hardcoded
State persistence	All component state written to disk atomically — survives restarts
Position sync independent	Runs while DEMO session active regardless of loop state
Watchdog 300s + auto-restart	3 auto-restart attempts before alerting; never kills during order confirm

7. Environment Fitness System

Factor	Weight & Scoring
ATR vs 20-period rolling average	30% weight — $< 1.2x = 100\text{pts}$ / $> 1.8x = 0\text{pts}$
15m trend strength	25% weight — Strong aligned = 100pts / Flat = 0pts
Session timing	20% weight — First 2hrs = 100pts / Final 30min = 20pts
Spread vs normal (dynamic median)	15% weight — $< 1.3x = 100\text{pts}$ / $> 2x = 0\text{pts}$. Gate 5 uses rolling median of last 200 spreads from MarketDataHub (A8).
Recent signal quality	10% weight — Last 5 clean = 100pts / Choppy = 0pts

Score Range	Behaviour
80–100 Excellent	Trade at full size
60–79 Good	Trade at standard size
40–59 Marginal	Trade at 0.75x size, ML threshold +2
20–39 Poor	WAIT — log reason
0–19 Hostile	WAIT — flag for review

Safe default: `environment_scorer.score()` returns 50 (neutral) during OHLC warmup (insufficient bars). `env_scorer_fallback` ERRORS are only raised for real failures after warmup completes — not on every tick during warmup.

8. Reliability Controls — Block A (A1–A12)

All A-series controls are committed and passing in the test suite (~232 tests). These are production behaviours, not optional enhancements.

ID	Control	Behaviour
A1	Margin preflight	Before live order: required = stop_pts × size × point_value × 2. Reduce size or skip if insufficient. Log 'Margin OK: £{avail}'. Fail silently on balance errors.
A2	Balance refresh	5 min with open positions, 30 min when flat → dashboard snapshot.
A3	Rapid drawdown	Rolling 60-min GBP loss > £100 → force WARNING for 30 min, auto-clear after.
A4	Pre-entry sync	Compare cached IG vs local position count; sync on mismatch; block if IG shows open.
A5	Market suspension	Market suspension codes → block orders for 5 min; show in gate panel.
A6	Stale pending trades	Open trades without IG deal > 24h → WARNING + ERRORS pill (pending_trade_stale).
A7	Watchdog banner	After 3rd real startup failure → src/data/logs/watchdog_failed.txt + permanent RED banner. Delete file + restart to dismiss.
A8	Dynamic spread median	Rolling median of last 200 spreads in MarketDataHub. Gate 5 compares current spread vs median (not static config).
A9	Directional cooldown	Cooldown key is epic:direction. BUY cooldown does NOT block SELL on same epic.
A10	Partial close confirm	On IG partial close: update local size, score banked pts, log 'PARTIAL CLOSE confirmed'.
A11	Min trades per setup	learning_min_trades_per_setup >= 10 (config: 10).
A12	Log rotation	engine.log: RotatingFileHandler 10 MB × 7. Do NOT rotate ml_training_store.jsonl (append-only forever).

Post-A Fixes (not in original block list)

Fix	Behaviour
Watchdog false count fix	Do not count benign duplicate-instance / already-running lock failures toward the 3-failure threshold. Reset failure count on successful startup. Clear false banners.
Environment scorer warmup	Insufficient bars during warmup → safe default fitness 50 without repeated env_scorer_fallback ERRORS pill each tick. Real failures still warn after warmup.

9. Historical Data Ingestion

9.1 Fetch Script

Property	Detail
Script	scripts/fetch_historical_ohlc.py
Source	IG REST API — paginated 5-minute bars
Cache output	src/data/ohlc_cache/nikkei_5m.jsonl (deduped, ordered append)
Resume behaviour	Resumes from last cached bar on restart
Page delay	12 seconds between pages to respect IG rate limits
Allowed window	07:00–22:30 Europe/London ONLY. Quiet outside that window.
Adaptive chunking	On IG exceeded-account-historical-data-allowance error: adaptive chunk sizing
Backoff	Exponential backoff on rate_limit / throttle errors
Graceful stop	Writes resumable state before stopping — can restart mid-range

9.2 Status File — src/data/state/ohlc_pull_status.json

Field	Purpose
last_success_timestamp	Timestamp of last successfully fetched bar
last_attempted_range	The date range attempted in the last fetch
block_reason	One of: allowance / rate_limit / throttle / other
next_retry_time	When the scheduler may next attempt a fetch
run_status	Current operational status of the fetch process

9.3 Fetch Scheduler

Property	Detail
Scheduler script	scripts/ohlc_fetch_scheduler.py
Window module	src/system/ohlc_fetch_window.py
Run mode	--once flag for cron, or loop mode
Window enforcement	Only runs inside 07:00–22:30; honours next_retry_time

Property	Detail
Agent coexistence	May run while live agent is active (uses OHLC REST budget window); shared budget still throttles under load

9.4 Known Limits — IG Historical Allowance

Demo and live accounts hit error.public-api.exceeded-account-historical-data-allowance. Fetch stops gracefully and resumes from last bar + status next_retry_time. This is NOT fixed by stopping the agent. Fetch window is 07:00–22:30 Europe/London only. Coexistence with live agent throttles under load due to shared REST budget.

10. Replay & Offline Analysis

10.1 Core Pipeline (B1–B3)

Step	Detail
B1 — Paginated OHLC fetch	IG REST paginated bars → src/data/ohlc_cache/nikkei_5m.jsonl
B2 — Walk-forward replay	100x replay through full signal engine → src/data/replay_results.jsonl
B3 — Analysis	Threshold/RSI/session analysis → src/data/replay_analysis.txt (appended)

10.2 Nightly Replay Scheduler (B4)

Property	Detail
Script	scripts/replay_scheduler.py
Schedule	Daily ~06:15 BST
Blackout	No run 22:30–07:00. Hard stop at 22:30.
Mode	Incremental replay + analysis append (not full re-run)
Classification	Operational subsystem — not required for live trading

Replay and the nightly scheduler are operational subsystems. The live trading engine runs independently. Replay enriches the ML model over time but its absence does not block live trading.

11. ML Pipeline (Optional — Disabled by Default)

USE_ML_SIGNAL: false by default. No live trading impact when off. All ML infrastructure is built and tested but gate 9 (ML gate) passes through when disabled.

11.1 ML Training Hooks (C1)

Item	Detail
File	src/trading/ml_training_hooks.py
record_entry	Called on IG confirm of entry — writes entry features to ml_training_store.jsonl
record_exit	Called on IG close — writes outcome label
Storage	src/data/ml_training_store.jsonl — append-only, NEVER rotated (A12)

11.2 Trade Autopsy (C2)

Item	Detail
Per-trade file	src/data/autopsy/{deal_id}.json
Index	src/data/trade_autopsy.jsonl (appended)
Trigger	On IG close confirmation — non-blocking
Rule	Autopsy only on IG-confirmed trades. Never on pending or phantom trades.

11.3 Training Dataset Builder (C3)

Item	Detail
Script	scripts/build_training_dataset.py
Inputs	replay_results.jsonl (1x weight) + ml_training_store.jsonl (3x weight)
Output	src/data/training_dataset.csv
Note	Real Demo labels always outweigh replay labels per section 3.2

11.4 Shadow Log (C4)

Item	Detail
File	src/data/shadow_log.jsonl
Written by	Every SignalEngine.evaluate() call
Fields	timestamp, scores, RSI, ATR, session, setup, would_have_fired
Purpose	Validates ML model on live ticks without gating real trades

11.5 ML Scorer (C5)

Item	Detail
File	src/trading/ml_scorer.py
Model	XGBoost (stub until trained)
Gate 9 blend (when enabled)	$(rules \times 0.6) + (ml_prob \times 100 \times 0.4)$
Enable flag	USE_ML_SIGNAL: false in config
Safe default on failure	Return 0.0 — treat as suppressed, log SAFE MODE

12. Web Dashboard — Complete GUI Rebuild

12.1 Technology Stack

Component	Technology
Backend API	FastAPI (Python) — same language as trading engine
Real-time data	WebSocket — push prices, trades, signals to browser instantly
Frontend	React — component-based, high-performance rendering
Charts	Recharts or TradingView Lightweight Charts
Styling	Tailwind CSS — dark professional theme
State management	React hooks — clean real-time state updates
Mobile	Responsive design — works on phone and tablet
Local access	http://localhost:8080
Remote access	Optional: ngrok or Tailscale for secure remote access

12.2 Dashboard Layout — Five Panels

Panel	Contents
LIVE	Large bid/offer price, IG STREAM status, agent state banner, active trade panel, Why No Trade, last signal confidence gauge, sentiment factor
TRADES	Active positions with live P&L, stop/target levels, trail status, partial close status. Closed trades in GBP with WIN/LOSS/Pending. IG-confirmed only.
POINTS	Live trade/session/cumulative points score, agent state, win rate chart, P&L curve, benchmark comparisons
INTELLIGENCE (NEW)	Replay summary, signal quality (shadow log), learning status (ML records, DB trades, top setups, progress to 500). Trade autopsy summaries, pattern library, weekly report.
SYSTEM	REST budget chart, stream health, position sync status, model version, last retrain, E2E test runner, emergency stop button, watchdog RED banner when watchdog_failed.txt exists

12.3 Intelligence Tab APIs (Block D)

Endpoint	Purpose
GET /api/replay/summary	Latest replay results summary

Endpoint	Purpose
GET /api/shadow/today	Today's shadow log signal quality stats
GET /api/learning/status	ML records count, DB trades, top setups, progress to 500
POST /api/replay/run	Trigger manual replay run (respects live window — blocked 22:30–07:00)

12.4 Real-Time Barometers — Always Visible Header

Barometer	Shows / Updates
BID / OFFER	Live Japan 225 price — large, prominent. Every Lightstreamer tick.
Agent State	HEALTHY / CAUTION / WARNING / DANGER / STOP. On points change.
Points: Trade / Session / Cumulative	Last trade score, today's total, all-time master score. After each close.
ML Confidence	Current ML score (0-1) gauge. Every signal.
Signal Strength	Current confidence (0–99) gauge. Every tick.
Environment Fitness	Fitness score + regime label. Every 5m bar.
Win Rate (Today + All Time)	Won/total % — updated after each close.
Daily P&L	Running GBP P&L. Every position update.
Stream Status	LIVE / STALE / DISCONNECTED. Every tick.
REST Budget	X/6 calls/min. Every REST call.
Spread current vs normal	Current spread vs dynamic median (A8). Every tick.
Sentiment factor	Client sentiment crowded_long / crowded_short flag (Block E). On session open.
Watchdog banner	Permanent RED banner if watchdog_failed.txt exists. Manual clear required.

13. Client Sentiment (Block E)

On each session open: GET /clientsentiment/{epic} called once and cached for the session.

Condition	Effect
80%+ long → crowded_long	BUY fitness penalty -10. Exposed via get_factors() and dashboard snapshot.
< 20% long → crowded_short	SELL fitness penalty -10. Same exposure.
API errors	Fail silently — sentiment factor defaults to neutral (0 adjustment).
Cache lifetime	Session duration — refreshed on next session open.

14. Session & Points State Persistence (Block F)

Item	Detail
Session end summary	write_session_end_summary called from trading loop on flatten. Unchanged path, verified.
points_state.json	Persists state and cumulative_points. Aliases to existing cumulative/nominal logic.

15. Complete System Architecture

15.1 All Components

Component	Responsibility
IG REST / Lightstreamer	Live tick reception, order routing, position sync — v24 proven
FastAPI server	Exposes all bot state via REST + WebSocket to web dashboard (read-only — NEVER writes to trading state)
Web Dashboard	React frontend — real-time barometers, trades, points, intelligence
Replay Scheduler	Nightly 06:15–22:30 BST — pulls bars, runs 100x replay, retrains
Replay Engine	Feeds historic bars through full agent at 100x — extends fast_replay.py
State Manager	Persists all component state atomically — survives restarts
Session Manager	Auto open/close handling, gap protection, cold start, DST times
Environment Scorer	5-factor fitness + regime + dynamic spread median + historical context
Points Engine	Trade/session/cumulative scoring, backing-off, recovery
ML Scorer	XGBoost via subprocess — scores signal at gate time (disabled by default)
Trade Manager	ATR trail, break-even, partial close, momentum exit, re-entry
Correlation Guard	Direction limits, confidence decay, intra-day win rate protection
Trade Autopsy	Analyses every confirmed close — pattern extraction
Insight Store	Pattern library — winning and failure conditions
ML Training Hooks	record_entry / record_exit on IG confirm — writes ml_training_store.jsonl
Shadow Logger	Every SignalEngine.evaluate() → shadow_log.jsonl
Client Sentiment	Session-open GET /clientsentiment/{epic} — crowded long/short fitness penalty
OHLC Fetch Scheduler	scripts/ohlc_fetch_scheduler.py — historical bar pull, 07:00–22:30 only
Benchmarking	5 continuous benchmarks — proves AI adding value
Reconciliation Script	scripts/reconcile_pending_trades.py — runs after each session
Emergency Stop	scripts/emergency_stop.sh — closes all + locks agent

15.2 The 13-Gate Decision Flow

Gate	Name	Pass Condition / Fail Action
1	Market open	Session hours OK, weekend block clear → else WAIT

Gate	Name	Pass Condition / Fail Action
2	Cold start	6+ complete bars since session open → else WAIT (log bars remaining)
3	Gap check	No extreme gap — or wait period elapsed → else WAIT (log gap size)
4	Environment fitness	Score ≥ 40 → else WAIT (log score and factors)
5	Points state	Not STOP, not session/day pause → else WAIT (log points state). Includes market suspension check (A5) and margin preflight (A1).
6	Correlation guard	Same-direction < 2 , signal not decayed → else WAIT (log reason)
7	Signal confidence	Confidence $\geq$ threshold (92 production) → else WAIT
8	Risk validation	Spread (dynamic median A8), ATR, position limits, GBP 100 cap → else WAIT
9	ML gate	ML score $\geq$ ml_threshold (or USE_ML_SIGNAL=false bypasses) → else WAIT (log ML score)
10	Size calculation	Points + ML + fitness → multiplier. Default 0.5x if error.
11	Execute (async)	IG order background worker — loop continues. Log failure.
12	Trail monitoring	Updates every 5m bar, partial close at 1.5x ATR. Log if fails.
13	Exit / session end	Trail, time, momentum, or session end flatten. Force close at T-5min.

16. Data Files & Operator Artefacts

Path	Purpose
src/data/ohlc_cache/nikkei_5m.jsonl	Historical 5m bar cache (deduped, ordered append)
src/data/state/ohlc_pull_status.json	Fetch scheduler / retry state
src/data/state/points_state.json	Points at all 3 levels, state, consecutive counts — updated after every trade
src/data/state/trail_state.json	Trail level, break-even, partial close per position — every 5m bar
src/data/state/session_state.json	Session open time, bars elapsed, gap detected — every 30s
src/data/state/replay_state.json	Last replay timestamp, bar count, calibration factor — after each replay
src/data/replay_results.jsonl	Replay signal outcomes
src/data/replay_analysis.txt	Threshold/RSI/session report (appended)
src/data/shadow_log.jsonl	Every evaluation shadow record
src/data/ml_training_store.jsonl	ML training rows (append-only, NEVER rotate)
src/data/autopsy/{deal_id}.json	Per-trade autopsy file
src/data/trade_autopsy.jsonl	Autopsy index (appended)
src/data/training_dataset.csv	Built training matrix (script output)
src/data/ml_model/	ML model pickle + meta (when trained). New file per version — never overwrite.
src/data/logs/watchdog_failed.txt	Persistent failure banner trigger. Delete + restart to clear.
src/data/logs/watchdog_restart_failures.json	Failure counter (cleared on successful startup)
src/data/logs/engine.log	Rotating 10 MB × 7 (A12)

17. Configuration

Key	Value & Notes
USE_ML_SIGNAL	false — ML gate inactive. Set true only after model trained and Demo-approved.
learning_min_trades_per_setup	10 — minimum trades before a setup influences model (A11).
signal_threshold (Japan 225)	92 — review after 20 sessions.
signal_threshold (EUR/USD)	88
signal_threshold (Wall Street)	90
signal_threshold (Gold)	90

18. Operations

18.1 Log Rotation

File	Policy
engine.log	RotatingFileHandler: 10 MB per file, 7 backups (A12)
ml_training_store.jsonl	Never rotated — append-only forever (A12)
shadow_log.jsonl	Append-only — archive manually when large
trade_autopsy.jsonl	Append-only index

18.2 Cron / Scheduler Examples

ohl_c_fetch_scheduler.py in --once mode (for cron):

```
# Run once from cron — window check is internal

30 7 * * 1-5 cd /path/to/ig_agent && python scripts/ohl_c_fetch_scheduler.py --once
```

Replay scheduler runs automatically at 06:15 BST — no additional cron required if replay_scheduler.py is running as a daemon.

18.3 Watchdog Banner Clearance

Item	Detail
Trigger	3rd real startup failure writes watchdog_failed.txt
Symptom	Permanent RED banner in SYSTEM panel
Clear procedure	1. Investigate engine.log for root cause. 2. Delete src/data/logs/watchdog_failed.txt manually. 3. Restart agent. Banner clears on successful startup.
Important	Benign duplicate-instance / already-running lock failures do NOT count toward the 3-failure threshold (post-A fix).

19. Multi-Instrument Architecture

v25 trades 4 instruments simultaneously with a single global position slot. Slice 1 launches with Japan 225 only. Additional instruments activated one at a time after Demo validation per instrument.

19.1 Instrument Registry

Instrument	IG Epic	Threshold	Best Session	Notes
EUR/USD	CS.D.EURUSD.CFD.IP	88%	12:00–17:00 BST	FX, 24hr, tight spreads, high liquidity
Japan 225	IX.D.NIKKEI.IFM.IP	92%	23:00–06:00 BST	Index, proven in v24, Session 1 passed 26/05/2026
Wall Street	IX.D.DOW.IFM.IP	90%	13:30–21:00 BST	Index, US session, high volume, wider spreads
Gold	CS.D.CFDSILVER.CFD.IP	90%	08:00–21:00 BST	Commodity, safe haven, different volatility profile

19.2 Activation Sequence

Phase	Gate to Activate
Slice 1 (Week 1)	Japan 225 only. Architecture built and proven. Others defined but disabled.
Demo Week 5 + EUR/USD	Japan 225 win rate stable at 55%+ for 2 weeks AND points state HEALTHY.
Demo Week 6 + Wall Street	EUR/USD stable for 1 week AND combined points state still HEALTHY.
Demo Week 7 + Gold	Wall Street stable for 1 week AND all 3 active instruments combined points HEALTHY.

20. Eight-Week Delivery Plan

Weeks 1–4: Build

Sprint	Build Targets
Week 1 — Core AI + Session Engine	state_manager.py, points_engine.py, environment_scorer.py, trade_manager.py, session_manager.py, replay_engine.py, replay_scheduler.py, emergency_stop.sh, FastAPI skeleton, first overnight replay (~2,000 labels), wire all 13 gates, config validation.
Week 2 — Trade Intelligence + Learning	trade_autopsy.py, insight_store.py, historical context from insight store, reconcile_pending_trades.py, weekly intelligence report v1, replay with autopsy features (~4,000 labels).
Week 3 — Correlation and Protection	correlation_guard.py, enhanced gap/cold-start detection, full regime detection (TRENDING/CHOPPY/VOLATILE/RANGING), intra-day win rate protection, ~6,000 labels.
Week 4 — Web Dashboard	FastAPI server with all state, React LIVE + TRADES + POINTS + INTELLIGENCE + SYSTEM panels, all 13 barometers + sentiment + watchdog banner, ML decision log, sound alerts, mobile-responsive. Tkinter still running in parallel. ~8,000 labels.

Weeks 5–8: Demo Monitoring

Period	Activity
Every session	Agent trades every Japan 225 session (~20 sessions over 4 weeks). Nightly replay continues. Reconciliation script runs after every session.
Weekly	Intelligence report reviewed every Sunday before market open. Model retrains weekly with approval. Real labels dominate by Week 6.
Week 6 mid-review	Win rate trend, points trajectory, benchmark status. New model staged for approval — GUI notification if unapproved > 24hrs. Auto-rollback: win rate < 50% for 3 days → revert to previous model.
Week 8 decision	All live gates passed? → Proceed to Live. Not passed? → Extend 2 weeks. NEVER rush Live.

21. Live Trading Gate — 16 Hard Gates

Every item below must be confirmed before allow_live_trading is set to true. This is not advisory.

Gate	Target
Demo trades	200+ complete cycles (open to close)
Win rate	60%+ over last 100 Demo trades consistently
Cumulative points	HEALTHY for 2+ consecutive weeks
Demo sessions	15+ sessions completed and logged
Open market proof	PASSED (v24 proven 25/05/2026)
ML contribution	ML adding > 5% win rate vs rules-only
Trail performance	ATR trails demonstrably improving P&L
Replay calibration	Within 10% of actual Demo performance
Trade autopsy	Running on every confirmed close — no gaps
Weekly retrain	Approval workflow tested 2+ complete cycles
Protective stops	Always sent — zero exceptions
Session end flatten	Zero positions held through session end
Emergency stop	Script tested — closes all and locks
State persistence	Restart mid-session tested — full recovery
Infrastructure	caffeinate + launchd stable 2+ weeks
Human sign-off	You are personally satisfied — judgement overrides checklist

22. Component Safety and Performance

22.1 Safe Defaults on Failure

Component	Safe Default
environment_scorer.score()	Return 50 — neutral, trading continues (warmup: always; real failures: warn after warmup)
points_engine.get_state()	Return HEALTHY — never block on points error
points_engine.get_size_multiplier()	Return 0.5x — conservative but allows trading
ml_scorer.score()	Return 0.0 — treat as suppressed, log SAFE MODE
trade_manager.update_trail()	Keep existing stop — never remove on error
session_manager.check_session_end()	Return False — safer than false positive close
FastAPI server failure	Trading continues headless — web dashboard reconnects
Replay scheduler failure	Log error, skip night — next night retries
Client sentiment API error	Fail silently — sentiment factor defaults to neutral

22.2 Performance Budget

Component	Max Time / Rule
All trading gates combined	< 600ms per tick — well within 5s loop
ML scorer subprocess	< 300ms — monitored
FastAPI WebSocket push	Async — never in trading path
Web dashboard render	In browser process — zero impact on trading
Replay (nightly)	Separate process — hard cutoff 22:30 BST
Trade autopsy	Runs post-close — never in tick loop
OHLC fetch	Separate process — 07:00–22:30 window only

23. Operational Safety

23.1 Emergency Stop

Step	Action
Run	bash scripts/emergency_stop.sh
Step 1	Close all open IG positions via REST API
Step 2	Set allow_live_trading = false in config
Step 3	Kill all agent processes
Step 4	Write emergency_stop.lock to project root
Recovery	Delete lock file manually, review config, restart
On startup with lock	Agent refuses to start — requires manual deletion

23.2 Multi-Instance Protection

On startup: check lock file AND running processes by pattern. If live instance detected: log error and refuse to start. launchd auto-restart increases this risk — protection is mandatory.

24. Absolute Rules — Non-Negotiable

- v24 main branch NEVER touched. All work on feature/v25-ml-enhancement.
- 14/14 E2E + 286/286 battery pass before EVERY commit — zero tolerance.
- Web dashboard (FastAPI + React) in Slice 4 — never extend Tkinter.
- USE_ML_SIGNAL = false until Slice 1 fully tested and Demo approved.
- Risk rules ALWAYS execute first — before fitness, points, ML gate.
- Trail ONLY moves in profit direction — never backwards.
- Session end close is mandatory — no positions through session end. Ever.
- Every component: try-catch, safe default, log warning, never crash.
- Replay labels 1x weight. Real Demo labels 3x weight. Never equal.
- Every model version is a new file — signal_model_v1.pkl, v2, v3. Never overwrite.
- State files written atomically — temp file then rename. Never corrupt on crash.
- Replay stops 22:30 BST hard cutoff — live trading always gets full REST budget.
- FastAPI server NEVER writes to trading state — reads only. Zero trading path impact.
- Tkinter removed in Slice 4 only — after web dashboard proven over 1 week live.
- Report after each numbered step in Slice 1 order. Await confirmation before next.
- All closed trade autopsy only on IG-confirmed trades — never on pending.
- Deal reconciliation: always store both open_deal_id and ig_close_deal_id.
- reconcile_pending_trades.py runs after every session as standard practice.
- OHLC historical fetch only within 07:00–22:30 Europe/London window — never outside.
- ml_training_store.jsonl is NEVER rotated — append-only forever.
- Watchdog banner requires manual deletion of watchdog_failed.txt + restart to clear.
- Benign duplicate-instance / already-running lock failures do NOT count toward watchdog failure threshold.
- Client sentiment fetch is once per session open — never intra-session.
- Replay scheduler and OHLC fetch are operational subsystems — NOT required for live trading.

25. Current Codebase State

Item	Status
Branch	feature/v25-ml-enhancement
v24-stable tag	cc0d522 — original stable
v24-progui-stable tag	0f7bee2 — proven trading agent (25/05/2026)
Last Phase 1 commit	c7edf51 — all ML scaffold + display fixes
GUI fixes committed	6d5b7ac, 520eef8, 49b0eb7, 904cde1
Blocks A–F commits	4648ba3–37e8e43 (reliability, ML, dashboard, sentiment, session)
Fetch hardening	3c7693e — fetch harden; fd140a6 — fetch schedule
Replay analyse	e4465cb — replay analyse endpoint
Checkpoint commit	7b27130 — chore: checkpoint all pending changes (may bundle unrelated workspace files — reference code paths in this spec, not every file in that commit)
Reconciliation script	scripts/reconcile_pending_trades.py
E2E tests	14/14 passing
Battery + soak	~232 passed (expanded from 286 with new test modules)
Signal threshold	92 in config — reviewed after 20 sessions
Lightstreamer	CONNECTED:WS-STREAMING — proven working
Open market proof	PASSED — trades_opened=1 closed=1 alignment=100%
Tkinter GUI	Replaced in Slice 4 with web dashboard

IG Agent v25 — Complete Final Specification v7 — Confidential

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